AM 6

ARMY MARS NET PLAN

January 2021

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DEPARTMENT OF THE ARMY
MILITARY AUXILIARY RADIO SYSTEM
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Page	Date / Change	Page
	June 2020. Major update. Incorporated content from several discontinued manuals and annexes	

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A Change 1		US Army

2. RETAIN THIS NOTICE AND INSERT BEFORE TABLE OF CONTENTS.
3. Holders of AM-6 will verify that page changes and additions indicated above have been entered. This notice page will be retained as a check sheet. This issuance, together with appended pages, is a separate publication. Each notice is to be retained by the stocking points until the standard is completely revised or canceled

i. Improvements

Chief Army MARS is the proponent of this publication. This manual is prepared at HQ Army MARS.

Suggested corrections, or changes to this document, should be submitted through the State Director to the Regional Director who will pass on a recommendation to HQ Army MARS.

ii. References:

The following references apply to this manual:

1. DoDI 4650.02 – Military Auxiliary Radio System
2. AR 25-6 - Military Auxiliary Radio System (MARS) and Amateur Radio Program
3. AR 25-1 – Army Knowledge Management and Information Technology
4. AR 5–12 – Army Management of the Electromagnetic Spectrum
5. Combined Communications Electronics Board, Allied Communication Publications.

ACP-121	Communications Instructions – General
ACP-125	Radio Telephone Procedures

6. Manual of Regulations and Procedures for Federal Radio Frequency Management
7. ATP 6-02.53 – Techniques for Tactical Radio Operations December 2019
8. Army MARS Documents:

AM 6 TTP 60 Meter Operations (U)

TABLE OF CONTENTS

TABLE OF FIGURES

1 GENERAL OVERVIEW

This Net Plan describes Army MARS (AMARS) communications support processing unclassified traffic, training, and daily MARS operations. The AM-6 NETPLAN replaces all other previous versions of the NETPLAN and its annexes, and is superior to all Region orders, plans and manuals.

> *Refer to the Table of Contents for all valid annexes at time of publication. Previously released annexes not listed in the Table of Contents are superseded.*

All AMARS on-the-air activity is governed by this NETPLAN and its annexes. The intent of this plan is to outline how contingency telecommunications support to DoD missions is technically implemented in the cyber-impaired or denied environment.

2 CONCEPT OF OPERATIONS

2.1 High Frequency (HF) Radio Concept of Operations

High Frequency (HF) Radio Networks span a 24-hour period although personnel are not scheduled or expected to guard the network continuously.

This plan describes a multi-layered radio network design and includes Wide Area (WAN) and Local Area Network (LAN) topologies. The following discussion is intentionally intended to describe MARS network topology using industry terminology.

2.1.1 The HF Local Area Network

The Local Area Network (LAN) is a network topology where all the stations in the network can communicate directly with all the other stations in the network. The LAN typically uses one frequency, and access to the net is governed by a Multiple Access Control (MAC) technique. Only one station can transmit at a time. HF LAN topology can be used in voice and data networks.

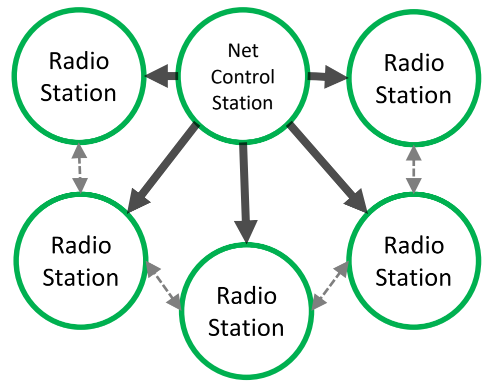

Figure 1 Local Area Network Topology

In a HF radio LAN, the network's defined Area of Operation (AO) is limited by a HF radio circuit that covers the AO so that all stations can communicate with all other stations in the net (see figure 1). In a larger AO, more than one frequency may be required to overcome coverage problems associated with mixed path distances. Such a circuit is a balance between planning a realistic AO, good frequency selection for all hours of the day, and enforcing Quality of Service (QOS)/radio station performance standards.

Collision avoidance is achieved through Carrier Sense Multiple Access (CSMA) techniques such as Listen-Before-Transmit (LBT). CSMA/LBT is the only means of collision avoidance in a Free Net. When the activity in the network increases, additional Access Control is required. A Net Control Station (NCS) functions similarly to a LAN Switch and directs traffic. This is called a Directed Net. Although Directed Net provides more control, Free Net is preferred as it is fast and utilizes the least amount of network overhead. Directed Net is only utilized when the additional overhead is required to manage the network to avoid collisions. Free Net is the normal status at all times, except when Directed Net is declared by the NCS.

2.1.2 The HF Wide Area Network

A Wide Area Network (WAN) is a network defined by a point to point topology between nodes. All stations in a Wide Area Network are generally not able to communicate directly with each other, rather, traffic is relayed through nodes according to a routing plan. In figure 2, in order for station A to pass a message to Station B, Station A must use a relay station.

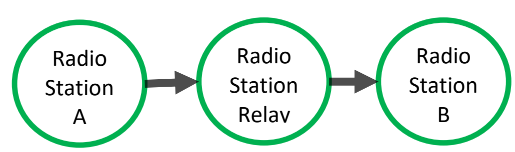

Figure 2 Wide Area Network Topology

Most long distance HF radio networks usually operate using WAN topology since communications over variable distances usually requires utilizing different radio frequency ranges. Small groups of powerful stations that have superior Radio Frequency (RF) capabilities can usually maintain communications on a common frequency for a period of time when propagation conditions are favorable. This technique is limited to the most powerful and well-developed of radio stations. Usually, an HF WAN consists of radio stations designated as “nodes”, “relay stations” or “gateways”. To achieve superior performance, ALE scanning equipment is utilized. Station to Station calls are made on a frequency trunk that spans several frequency bands throughout the HF spectrum.

2.1.3 Mesh concept in the HF LAN

A mesh network is LAN topology in which the nodes interact directly, dynamically and non-hierarchically to as many other nodes as possible and cooperate with one another to efficiently route data from/to clients. The concept is applied to HF radio networking using manned stations. Multiple stations work together and form a pseudo “Mesh Network” significantly reducing the requirement of weaker stations to retransmit messages. All stations are responsible for receiving all traffic. When a re-transmission is required from a weak station, the weak station is not asked to retransmit, rather, the strongest station in the net that received the traffic is used to retransmit the message (see figure 3).

The technique is very efficient when operating in Free Net, as coordination through the NCS tends to decrease message throughput speed.

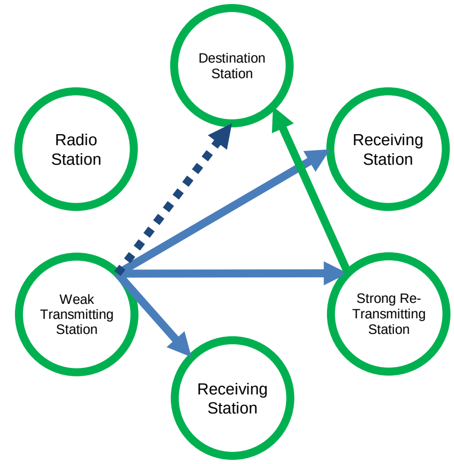

Figure 3 HF Mesh Relay in Free Net

2.1.4 Automatic Link Establishment (ALE) in the HF WAN

ALE is best suited as a One-to-One communications link, although One-to-Many calls are possible when all stations are defined in a limited Area of Operation that supports communications on a single radio frequency (see figure 4).

Traffic nodes for Tributary Station (Sub-Region Level), Minor Relay Station (Region Level), Intermediate, Major and Primary Relay Stations (National Level) use unique ALE addresses assigned to their role.

2.1.5 Continuous Monitoring

Continuous Monitoring refers to stations being tasked to monitor established calling frequencies on a continuous basis, up to 24 hours daily. The Continuous Monitoring condition presents significant difficulties for a volunteer force to sustain and to maintain and protect electronic equipment in an austere environment.

Continuous Monitoring for All-MARS is only directed at the HQ level to support national requirements in the most extreme circumstances. Continuous Monitoring may also be utilized in a limited manner, applied to specific MARS stations and limited to a specific coverage area as required supporting an exercise or operation.

Calling frequencies for Continuous Monitoring are listed in AM 6 NETPLAN Annex D, included in MSC software.

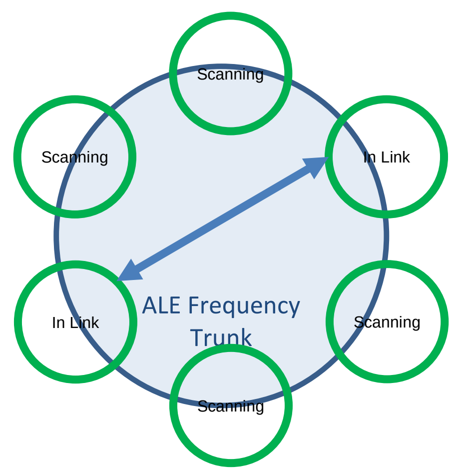

Figure 4 ALE Point to Point Topology

2.1.6 Periodic Monitoring

Periodic Monitoring is the normal, sustainable condition in which MARS stations normally operate in ordinary time, exercises, operations, and emergencies. Specific “operating windows” are established in each Region and referred to as the Sustained Network Operating Plan (SNOP). This condition defines time periods in which no MARS activity occurs to conserve resources, to allow electronic equipment to be placed in a protected condition, and allows MARS members to focus the majority of their available time on family, sustainment and employment.

Calling frequencies for Periodic Monitoring are listed in the AM 6 NETPLAN Annex D, which is included in MSC software.

2.2 Very High Frequency (VHF) Radio Concept of Operations

VHF radio is utilized between stations located in the same or nearby communities to establish situational awareness, pass traffic between nearby stations operating on different HF networks, and coordinate volunteer human resources in the local area. In general, only simplex and limited repeater operations are desired. It is the policy of AMARS to:

- Minimize the use of VHF
- Minimize infrastructure and member costs associated with operating and maintaining Land Mobile Radio Systems (LMR) utilized in MARS
- Utilize antenna systems and transmitter power to maximize the utility of common simplex channels whenever possible (see para 3.1.3 and 3.1.4)
- When data is required, utilize the same channels assigned to voice operation
- Conform to established technical and interoperability standards
- Avoid duplicating services available through the Amateur Radio Service (ARS), the Private Land Mobile Radio Service (PLMRS) and the Army LMR networks
- Utilize VHF in a manner that conserves radio frequency resources.

3 Technical Requirements

The AMARS network is expected to respond dynamically, demonstrate flexibility in adapting to new requirements, to be interoperable with US Military, and be suitable for volunteer personnel with limited time and material resources. To achieve this objective, AMARS implements a very narrow technical focus that ensures interoperability with US military and avoids proprietary and nonstandard technologies.

3.1.1 Quality Of Service (QoS)

The desired circuit quality for stations is "Good Readable" radio-telephone communications. To the extent that it is possible, all aspects of station design should be optimized to achieve this circuit quality through the majority of anticipated communications conditions over intermediate to long distances.

Stations that do not meet the desired quality of service compromise overall network efficiency, throughput, and create an unnecessary burden for other network stations.

3.1.2 Regulations

The National Telecommunications and Information Administration (NTIA) is the government authority managing Federal use of radio spectrum, including all stations in MARS service.

Notice of Apparent Liability
The FCC regulations and provisions of the NTIA Manual are found in 47 CFR 300. Compliance with FCC and NTIA provisions, rules and technical standards are mandatory. In general, punitive measures resulting from violations of the CFR are applicable to intentional and unintentional instances of noncompliance with FCC rules and the NTIA Manual. Refer to 47 USC 503 regarding punishment of any person who willfully or repeatedly fails to comply with any of the provisions stated. (Ref AR 5-12, sec. 6-4)

3.1.3 Permissible Radio Equipment and Emissions

Radio equipment with 47 Code of Federal Regulation (CFR) Part 80 or Part 90 Federal Communications Commission (FCC) type acceptances, Joint Interoperability Test Command (JITC) certification, or military equipment carrying a US nomenclature in current use by US Forces may be used by MARS members.

- 1.6 to 30 MHz: MARS members MAY use modified FCC Part 97 type accepted High Frequency (HF) Amateur Radio equipment providing the operation of the Amateur Radio equipment meets technical specifications as defined in the NTIA Manual of Regulations section 7.3.9.
- Only licensed amateur radio operators are eligible to volunteer for MARS membership. Government owned stations are not MARS stations and subject to the technical standards in the NTIA Manual of Regulations chapter 5, and therefore MAY NOT utilize Amateur Radio equipment when operating on AMARS frequencies in addition to requirements of the respective service Frequency Management Office and installation spectrum manager. There is no exemption for Amateur Radio equipment.

- 136-174 MHz: MARS stations are required to conform to NTIA Manual of Regulations Chapter 5, or FCC Part 90 when operating on government/military Very High Frequency (VHF) frequencies.
- MARS members may use existing Amateur VHF radio equipment that operates in Narrow mode (11K2F3E) through the end of its service life, or until 1 JAN 2023, whichever comes first. All new purchases of VHF equipment for use in MARS must be FCC Part 90 certified in order to comply with NTIA and DA regulations.
- MARS stations utilizing VHF may use Analog Voice (11K2F3E) or P25 (8K10F1E and 8K10F1D). MARS stations operating in "wide band" (20K or 25K) are in violation of NTIA and Army regulations and are subject to FCC enforcement action if not taken down. Limited use of packet radio (F2D) is permitted providing narrow band FM (11K2) is utilized.

3.1.4 Maximum Power Permitted

MARS stations are authorized to use up to the following transmitter power levels to maintain reliable communications.

Unless otherwise noted the maximum power for operation on AMARS HF frequencies is:

2-30 MHz--one kilowatt (1KW)

Unless otherwise noted, the maximum power for operation on AMARS VHF frequencies is:

136 to 174 MHz -- 100 watts maximum

3.1.5 National Radio Quiet Zone

Stations located within 90 miles of the Canadian or Mexican borders, or in the National Radio Quiet Zone (NRQZ) <u>may</u> have lower power limitations <u>on a case by case basis</u> if an interference condition exists. Those AMARS members affected will be notified individually of the specific restrictions that apply to them.

Figure 5 National Radio Quiet Zone

3.1.6 Requirements for ALE Stations

ALE utilized by MARS stations shall be according to MIL-STD 188-141B. MARS stations using ALE must be minimally capable of:

- Transmitting and receiving on all assigned ALE Trunk frequencies.
- Utilizing a broadband antennas or automatic tuning equipment that will tune the antenna in less than 2 seconds before transmitting.
- Scanning the trunk frequencies at a rate of at least 2 channels per second.

Stations that are not initially capable of meeting this minimum requirement, but have committed to meeting this requirement, may participate in training under the condition their limited capabilities are appropriate given the type of training being conducted, their participation does not detract from the training and operational objectives, and at the discretion of the relevant Region and/or Network Managers.

4 Network Operating Instructions

The following instructions apply to AMARS network operations:

1) General Instructions
   i. Brevity shall be practiced on all networks, using as few words as possible, to convey only relevant information, thoughts and ideas.
   ii. Abbreviated procedures shall be utilized whenever practical to do so.
   iii. MARS stations should be configured to meet the desired QoS for most conditions.
   iv. Stations should use as much power as necessary to reliably achieve the desired QoS for voice and data transmissions within the limits established in this plan. Antenna systems of sufficient size and height above ground should be used to achieve good communications quality.
   v. AMARS stations shall utilize the Serial Phase Shift Keying (PSK) or Frequency Shift Keying (FSK) waveforms described in MIL-STD 188-110D when text messaging is required.
   vi. Unless otherwise noted, Upper Side Band (USB) is utilized on all US military radio frequencies.
   vii. When it is necessary for MARS members to interface with Amateur Stations on ARS frequencies, or SHARES stations operating on SHARES frequencies, the prevailing mode and procedure in use will be utilized by the MARS station.

2) Network Operations
   i. Radio Telephone (voice) in according with ATP 6-02.53 is the primary operating mode on all networks. All stations operating in radio teletype (text) shall concurrently guard for voice calls and respond to those calls in voice mode.
   ii. Radio Telephone (voice) is the primary means of coordinating communications and traffic once a Radio Frequency (RF) communications link is established with ALE.
   iii. Radio networks are Free Nets unless declared a Directed Net by the NCS.
   iv. Net Call Signs are discontinued. Make net call up to "All Stations This Net".
   v. Radio networks operate for as long, or as little time as required. Close directed networks when business is complete. Do not hold networks open for late stragglers or for the purpose of a participation hour.
   vi. Radio frequencies should be stored in memory and frequency dials locked out to avoid inadvertent frequency changes.
   vii. Avoid network congestion by moving traffic from the calling frequency to a working frequency.
   viii. Assume all MARS stations are in possession of the frequency designator list and assume military and government stations do not have the list.
   ix. Sounding is limited to Primary, Major, and Intermediate Relay Stations and only on the assigned ALE trunk. All other stations are prohibited from sounding. LQA Calling (TWAS ALE Call on each channel) is the primary means to generate a Link Quality Analysis (LQA).

3) Messages
   i. Messages are sent by text using the HQs MARS provided MSC software, or by voice.
   ii. Messages originated by MARS members and stations shall only contain true and factual information. No MARS station may, under any circumstances including a training or exercise, draft or send a message that contains false, fictitious, sensualized or fantastical information. Transmitting fictitious information is grounds for dismissal per AR 25-6.

4) Security
   i. Text messages ***shall not*** be sent in the clear. When cryptographic capability exists at the transmitting and receiving station(s), MARS Type 3 on-line encryption, also known as Cipher Text (CT) Mode, is utilized to transmit text messages. HQ MARS will announce approved exceptions to this requirement.

      (Note the terms TRANSEC and COMSEC should not be used to describe MARS capabilities)

   ii. Text messages sent without on-line encryption shall use the CODRESS capability provided in MSC software. CODRESS messages should be sent with on-line encryption, and may be sent in the clear.
   iii. Use only approved cryptographic keys provided by HQ MARS. Expired or unapproved keys shall not be utilized.
   iv. Minimize all information and detail not specifically required or requested.
   v. State names or abbreviations shall not be sent along with call signs.

vi. The numerical frequency and frequency designator shall not be transmitted together while operating in Plain Text (PT) mode.
vii. MARS stations assigned a tactical call sign shall not also identify using their individual international or billet MARS call sign when the tactical call sign is in use.
viii. If issued, the authentication table for the current month will be used for operations. Cross out combinations that are known to be previously used or observed. The training table is used when it is anticipated that enough authentication combinations to compromise the month's table will be revealed in on-air training between known stations.

5) Training

i. On Air training activities are limited to exercising tasks, such as checking into nets, and sending and receiving messages. Training instruction, which includes describing in detail specific tactics, techniques and procedures, or discussion of matters of policy, shall be conducted off air utilizing mediums such as a telephone bridge. Radio nets should be conducted in parallel with training teleconferences so tasks can be practiced on air. This procedure improves the overall training benefit for all on air stations because propagation challenges are eliminated.
ii. Individual members in basic training shall utilize the call sign prefix AAT. When they have completed basic training their call sign prefix will change to AAR. The use of /T is discontinued.
iii. Training and familiarization with Amateur Radio techniques and operating modes will be through individually initiated Amateur Radio activities conducted on ARS frequencies. When working with the amateur radio community, it is permissible to announce that you are affiliated with MARS but you should not associate your MARS call sign with your amateur radio call sign.

4.1 Radio Station Call Signs

4.1.1 Assigned Call Signs

All AMARS radio stations are issued an internationally recognized call sign by the Chief Army MARS and it is printed on a MARS member's New Member Welcome Letter or Authorization To Operate (ATO). International Call Signs are required when MARS stations are operating in non-US Military networks, such as SHARES.

The following call sign prefixes are assigned by the Chief Army MARS to AMARS member stations:

AAA0AA through AAA9ZZ	Army MARS Billet Call Signs
AAR0AA through AAR9ZZ	Individual AMARS Member Stations Continental US (CONUS)
AAT0AA through AAT0ZZ	Individual AMARS Member Stations in Training CONUS
ACM3AA through ACM3ZZ	Individual AMARS Member Stations Virgin Islands
ACM4AA through ACM4ZZ	Individual AMARS Member Stations Puerto Rico
ALM7AA through ALM7ZZ	Individual AMARS Member Stations Alaska
ALM8AA through ALM8ZZ	Individual AMARS Member Stations Aleutian
ABM6AA through ALM6ZZ	Individual AMARS Member Stations Hawaii
ABM3AA through ALM3ZZ	Individual AMARS Member Stations Guam
AEM1AA through AEM1ZZ	Individual AMARS Member Stations Germany

FIRESTAR: A collective call sign for any Army MARS station that can establish radio communications. FIRESTAR is used for making a blind call to any MARS station.

4.1.2 Radio Appointment Titles

ACP-125 (G) describes Radio Appointment Titles, which are words used to identify a senior individual by role. They are not call signs, rather titles used to refer to a staff position or function in the same manner as billet call signs. For example, "REQUEST COMMS WITH MOONBEAM". The following appointment words are useful in MARS when referring to certain billeted officers or their designees during radio communications.

Title	Position
SUNRAY	Commander (Director)
MOONBEAM	Chief of Staff, Executive Officer (XO)
MANHOLE	Administrative Officer (S1) or staff
ACORN	Intelligence Officer (S2) or staff
SEAGULL	Operations Officer (S3) or staff
MOLAR	Logistics Officer (S4) or staff

(title) MINOR Used to designate the second-most senior appointment

4.1.3 Billet Call Signs

With the release of this update, billet call signs are authorized by the Chief Army MARS and are limited in scope to certain leadership and senior staff. The available billets are described in AM-1; however, not every staff position is assigned a Billet Call Sign. The specific billet Call Signs that are assigned to AMARS leaders are listed in the Global Address Book included in MSC software will be: Region Director, Executive Officer/Chief of Staff, Admin Officer, Operations Officer, State MARS Director, and State Admin Officer.

Examples:

AAA1RD	Region One Director
AAA1RX	Region One Executive Officer/Chief of Staff
AAA1R1	Region One Admin Officer
AAA1R3	Region One Operations Officer
AAA3PA	Army MARS Pennsylvania State MARS Director
AAA3PA1	Army MARS Pennsylvania Admin Officer

4.1.4 Temporary Mission Support Call Signs

Each Region Director has available a block of call signs to assign to radio stations on a temporary and limited basis. In an emergency situation, when life, property or quality of life is at risk, and administrative channels with Chief Army MARS or HQ Army MARS are severed, the Region Director may issue a temporary call sign for up to 72 hours to support a mission tasked by DoD. The call signs listed below may be assigned to radio stations of US military units and civil authorities that do not otherwise have call signs assigned by US military or US government authority such as the Army Frequency Management Office, NTIA or FCC, when operation on Global High Frequency Enterprise Radio Network (GHFERN) is required.

Region 1 AAT101-AAT112
Region 2 AAT201-AAT212
Region 3 AAT301-AAT312
Region 4 AAT401-AAT412
Region 5 AAT501-AAT512
Region 6 AAT601-AAT612
Region 7 AAT701-AAT712
Region 8 AAT801-AAT812
Region 9 AAT901-AAT912
Region 10 AAT001-AAT012

4.2 RADIO FREQUENCY PLAN

4.2.1 Authorized Radio Frequencies

MARS stations may operate in official AMARS activities on the radio frequencies listed in the AM 6 NETPLAN Annex D, which is included in the MSC software.

4.2.2 Calling Frequencies

Calling Frequencies are utilized for establishing communications with military, government and MARS stations and are listed according to priority in AM 6 NETPLAN Annex D.

4.2.3 ALE Trunk Frequencies and Configuration

The radio frequencies for the ALE trunks are listed in AM 6 NETPLAN Annex D. Three ALE frequency trunks are organized geographically to reduce loading and prevent congestion. Stations only scan the frequency trunk associated with their area. In order to establish communications with a station in another geographic region, call the desired station in their ALE trunk group (see figure 6).

Tune Time (seconds)	Scan Rate (channels per second)		Max Scan Channels (establishes sounding and scanning calling duration time)	Total observed Sounding or Scanning Call duration (seconds)	
2	When 10 or fewer channels are scanned When 11 or more channels are scanned	2 or 5 > 5	adjust for desired Sound/Calling Duration	2G ALE	10

Figure 6 ALE System Configuration

4.2.4 Long Haul Phone Patch Frequencies
Phone Patch Calling Frequencies are listed on AM 6 NETPLAN Annex D, and are exclusively for operation of the "*MARSRADIO*" Net. Only radio stations operated by active duty US Military, Reserve and National Guard, and those MARS stations who are specifically authorized by Chief Air Force MARS may use these radio frequencies.

4.2.5 Discrete (Working) Frequencies
Discrete Frequencies are shown in AM 6 NETPLAN Annex D. These MARS all-purpose working frequencies are utilized as required to prevent congestion on calling frequencies. During a disaster, the Region Discrete Frequencies may be prioritized for use in or near the affected area and minimized in other areas.

4.2.6 60 Meter Band Amateur Radio Channels
The five channels at 60 Meters which are assigned to the Amateur Radio Services on a secondary basis may be utilized by MARS stations for limited purposes. The conditions of their use are fully explained in AM6 TTP 60 Meter Operations. MARS members may only use their MARS call signs when designated by HQ MARS.

4.2.7 Canada/United States (CANUS) Radio Frequencies
Specific MARS and Canadian Forces Affiliate Radio Service (CFARS) frequencies are coordinated and authorized for operation in both countries. These radio frequencies are also used for ALE between stations in CA and the US and are listed in AM 6 NETPLAN Annex D.

4.2.8 OCONUS Radio Frequencies and Split Operation
In some cases, split-frequency operation will be necessary to communicate to an area outside of the Continental United States (OCONUS).

4.2.9 VHF Radio Frequencies
A nationally authorized VHF calling frequency is listed in AM 6 NETPLAN Annex D. This simplex frequency is for call up of MARS stations in all areas.

All repeaters should utilize a tone on both input and output frequencies. The standard tone frequency is 100.0 Hz. When a requirement exists for a repeater to operate on another tone, it should use the unique tone on the input and retain 100.0 Hz on the output.

All simplex voice operation should utilize CTCSS tone 100.0 Hz.

Packet operation should not use a CTCSS tone, unless required for repeater access.